DC Circuit Analysis, Power and Energy



Basic Electrical and Electronics Engineering

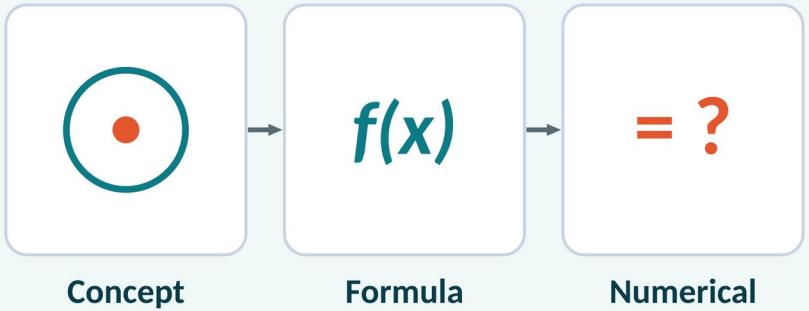
Unit 1: Circuit Fundamentals

Topic: Basic nodal and mesh analysis with independent DC sources; power and energy

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Learning Outcomes

- Apply KCL and KVL to DC networks.
- Solve circuits by nodal analysis.
- Solve circuits by mesh analysis.
- Calculate power and energy in resistive loads.

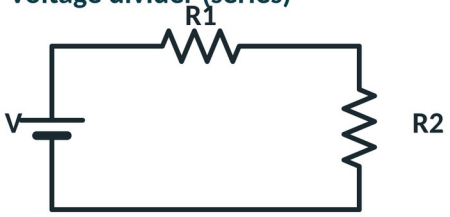


Linked to CO1
Analyse DC and AC electrical circuits to determine circuit parameters.

Recap: What You Already Know

- Ohm's law: V = I·R.
- Series: same current, voltages add.
- Parallel: same voltage, currents add.
- Equivalent resistance simplifies a network.

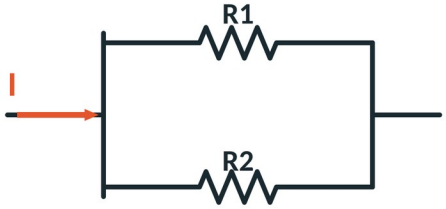
Voltage divider (series)



$V_2 = V \cdot R_2/(R_1 + R_2)$

$I = V/(R_1 + R_2)$

Current divider (parallel)



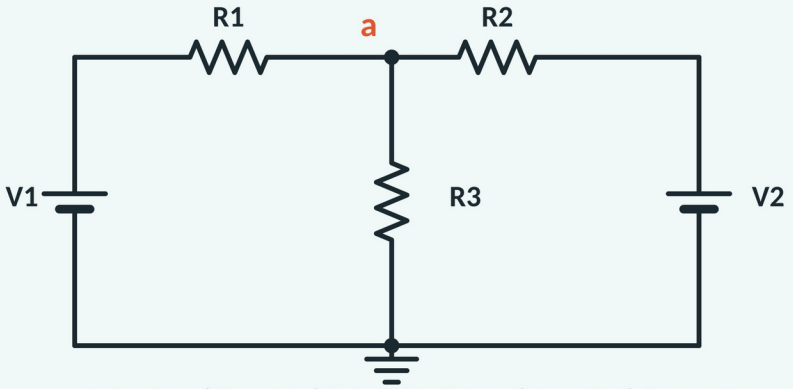
$I_1 = I \cdot R_2/(R_1 + R_2)$

$I_2 = I \cdot R_1/(R_1 + R_2)$

Series: $R_{eq} = R_1 + R_2$; parallel: $R_{eq} = R_1R_2/(R_1 + R_2)$
V volt, I ampere, R ohm

Motivation: Circuits with Many Sources

- Battery banks, solar panels and chargers share loads.
- Series–parallel reduction fails with two sources.
- Systematic methods always work.
- Same method used by circuit simulators.



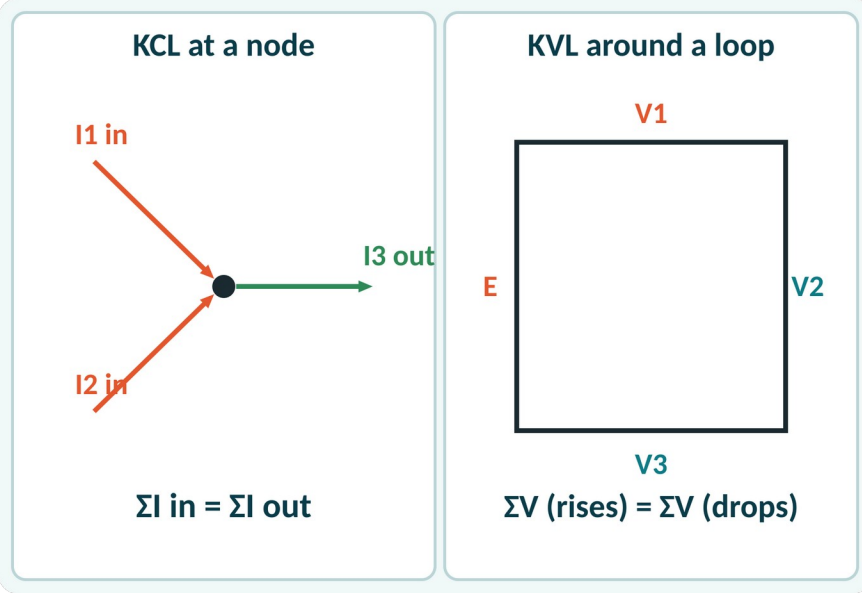
KCL at a: (Va − V1)/R1 + Va/R3 + (Va − V2)/R2 = 0

unknown: node voltage Va (ground = 0 V)

Systematic methods: nodal (KCL) and mesh (KVL)

Kirchhoff's Laws

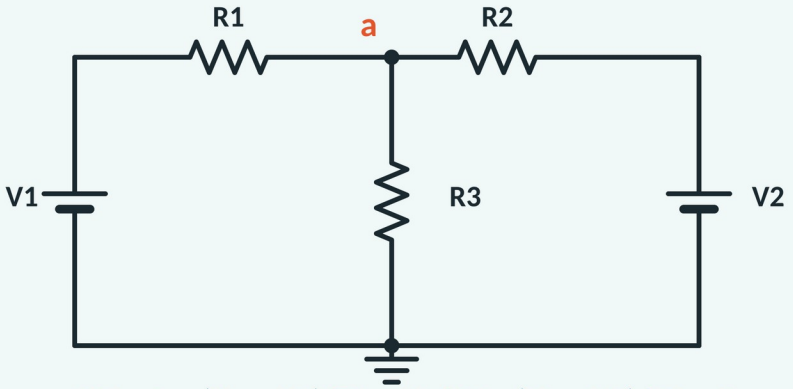
- KCL: current into a node equals current out.
- KVL: sum of voltage rises = sum of drops in a loop.
- KCL is conservation of charge.
- KVL is conservation of energy.
- Mark current directions first.



KCL: ΣI(in) = ΣI(out); KVL: ΣV = 0 around any closed loop

Nodal Analysis

- Choose a reference node (ground = 0 V).
- Label the other node voltages.
- Write KCL at each non-reference node.
- Express branch currents with Ohm's law.
- Solve the equations for node voltages.



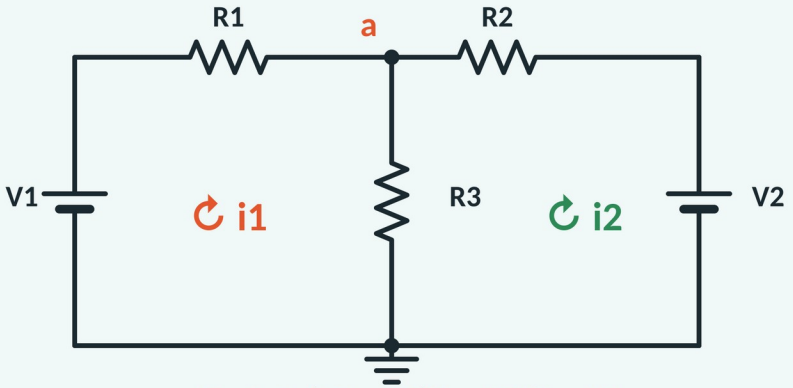
KCL at a: (Va − V1)/R1 + Va/R3 + (Va − V2)/R2 = 0

unknown: node voltage Va (ground = 0 V)

(Va − V1)/R1 + Va/R3 + (Va − V2)/R2 = 0
Number of equations = nodes − 1

Mesh Analysis

- Draw a clockwise mesh current in each loop.
- Write KVL around each mesh.
- Shared resistor carries the difference of currents.
- Solve the equations for mesh currents.
- Branch currents follow from the mesh currents.



Mesh 1: (R1 + R3)i1 − R3·i2 = V1

Mesh 2: −R3·i1 + (R2 + R3)i2 = −V2

(R1 + R3)i1 − R3·i2 = V1; −R3·i1 + (R2 + R3)i2 = −V2

Number of equations = number of meshes

Choosing the Method

- Count nodes and meshes first.
- Fewer equations means less work.
- Nodal suits current sources.
- Mesh suits voltage sources.
- Both give the same answer.

	Nodal	Mesh
Law used	KCL	KVL
Unknowns	node voltages	mesh currents
Choose when	fewer nodes than meshes	fewer meshes than nodes
Current sources	easy	need care
Voltage sources	need care	easy

Equations: nodal n − 1; mesh b − n + 1 (b branches, n nodes)

Power and Energy

- Power: rate of energy conversion.
- Resistor power turns into heat.
- Energy = power × time.
- Utility unit: kWh (one unit).

$P = V \cdot I$

watt (W)

$P = I^2 \cdot R$

heat in a resistor

$P = V^2/R$

when V is known

$W = P \cdot t$

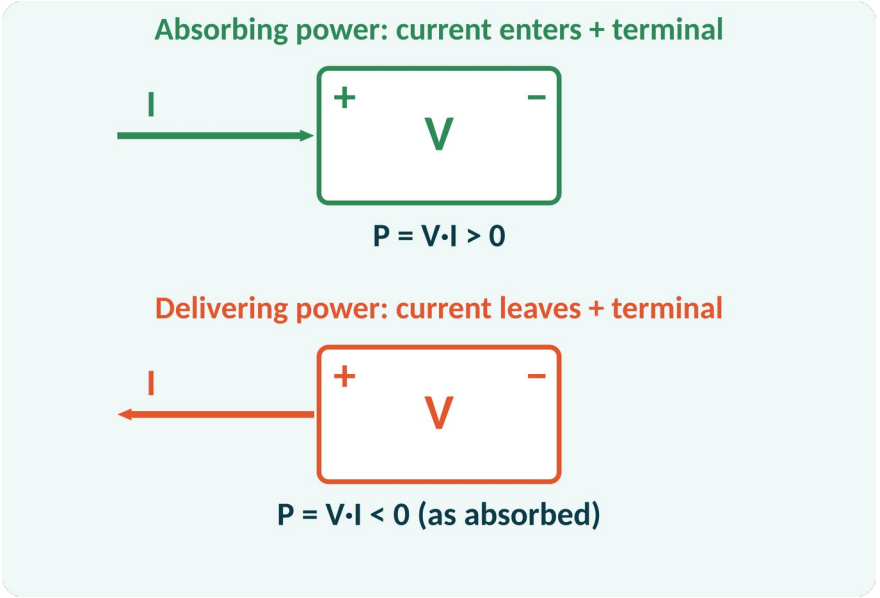
joule (J) or kWh

1 kWh = 3.6×10^6 J = 1 unit of electricity

$P = VI = I^2R = V^2/R$ (W); $W = P \cdot t$ (J, or kWh)
1 kWh = 3.6×10^6 J

Sign Convention for Power

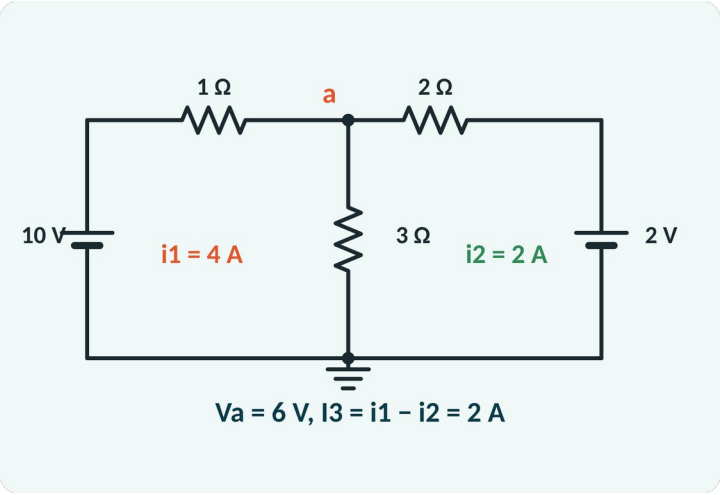
- Current into the + terminal: element absorbs.
- Current out of the + terminal: element delivers.
- Resistors always absorb.
- Total power delivered = total power absorbed.



Conservation: ΣP(delivered) = ΣP(absorbed)

Solved Example 1: Nodal and Mesh

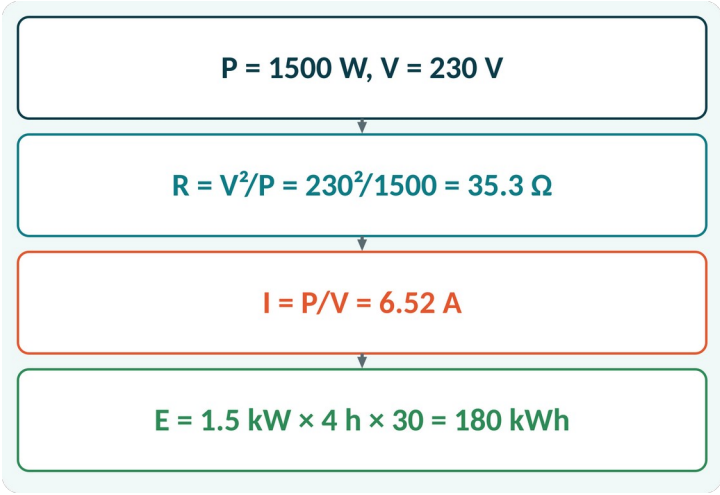
- V1 = 10 V, V2 = 2 V, R1 = 1 Ω, R2 = 2 Ω, R3 = 3 Ω (as drawn). Find Va, branch currents, power in each resistor.
- 1. Nodal: (Va − 10)/1 + Va/3 + (Va − 2)/2 = 0 → 11Va/6 = 11 → Va = 6 V.
- 2. I1 = (10 − 6)/1 = 4 A; I2 = (6 − 2)/2 = 2 A; I3 = 6/3 = 2 A.
- 3. Mesh check: 4i1 − 3i2 = 10 and −3i1 + 5i2 = −2 → i1 = 4 A, i2 = 2 A.
- 4. P(R1) = 16 W, P(R2) = 8 W, P(R3) = 12 W: total 36 W.



Answer: Va = 6 V; currents 4 A, 2 A, 2 A; V1 delivers 40 W, V2 absorbs 4 W, resistors dissipate 36 W

Solved Example 2: Heater Energy

- A heater is rated 1.5 kW, 230 V. Find R and I. If used 4 h per day for 30 days, find the energy and the bill at ₹6 per unit.
- 1. $R = V^2/P = 230^2/1500 = 35.3\ \Omega$.
- 2. $I = P/V = 1500/230 = 6.52$ A.
- 3. Energy per day = 1.5 kW × 4 h = 6 kWh.
- 4. For 30 days: 180 kWh.
- 5. Bill = 180 × ₹6 = ₹1080.



Answer: R = 35.3 Ω, I = 6.52 A, E = 180 kWh, bill = ₹1080

Common Mistakes and Exam Tips

Common mistakes

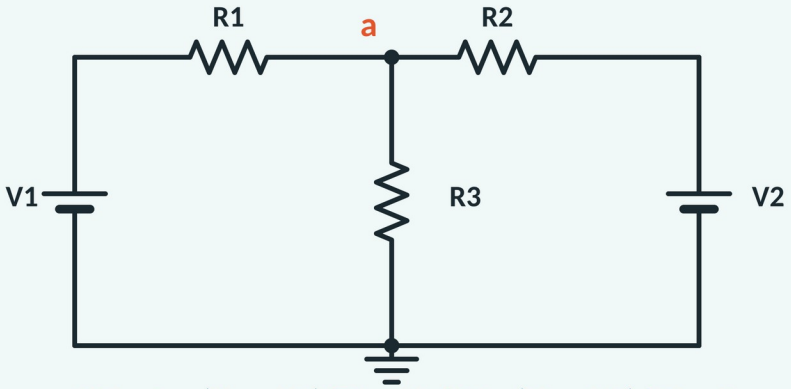
- Wrong sign for a source or current direction.
- Forgetting the shared-resistor term in mesh analysis.
- Writing too many or too few equations.
- Using P = V·I with V in mV and I in A.
- Mixing kW and W in energy calculations.

Exam tips

- Mark polarity and current directions first.
- Check power balance at the end.
- Count nodes (n − 1) or meshes before solving.
- Convert all units to SI.

Summary

- KCL: ΣI in = ΣI out; KVL: ΣV = 0 in a loop.
- Nodal: unknowns are node voltages; n − 1 equations.
- Mesh: unknowns are mesh currents; one equation per mesh.
- $P = VI = I^2R = V^2/R$; $W = P \cdot t$.
- Delivered power = absorbed power.



KCL at a: (Va − V1)/R1 + Va/R3 + (Va − V2)/R2 = 0

unknown: node voltage Va (ground = 0 V)

Choose the method with fewer equations; verify by power balance

Quick Quiz

Q1. Number of nodal equations for a circuit with 5 nodes:
(a) 3 (b) 4 (c) 5 (d) 6

Ans: (b)

Q2. Power in a 10 Ω resistor carrying 2 A:
(a) 20 W (b) 40 W (c) 5 W (d) 80 W

Ans: (b)

Q3. KVL expresses conservation of:
(a) charge (b) energy (c) mass (d) momentum

Ans: (b)

Q4. A 2 kW heater used for 3 hours consumes:
(a) 2 kWh (b) 3 kWh (c) 6 kWh (d) 6 kW

Ans: (c)

Q5. Current entering the + terminal means the element:
(a) delivers power (b) absorbs power (c) is open (d) is a short

Ans: (b)

References

1. **Hughes Edward, Electrical Technology**
   10th Ed., Pearson Education, 2010 (Units 1, 2 textbook)

2. **H. Cotton, Electrical Technology**
   7th Ed., CBS Publishers, 2005 (reference)

Next lecture: sinusoidal voltage, RMS and phasors (already prepared): BEEE_U1b